pitfalls can be corrected rather easily with statistical corrections such as Platt scaling. The core idea is to extremize the forecasts (Baron et al., 2014), correcting the output probabilities along a sigmoid curve. In practice, this process pushes low probabilities further down as they get closer to 0.0 and high probabilities higher up as they get closer to 1.0. We explore popular statistical calibration methods, and demonstrate that Platt scaling and a certain class of extremization techniques are mathematically equivalent. Overall, we see that Platt scaling is an essential statistical correction technique for achieving the best possible performance.[3]

We now turn to our primary results.

4 The AIA Forecaster Is Indistinguishable from Superforecasters

4.1 Benchmarks and Evaluation Metrics

We report the performance of the AIA Forecaster on three variants of the ForecastBench benchmark (Karger et al., 2024), as well as an additional benchmark (the MarketLiquid benchmark), which we describe below. Each of these benchmarks covers a diverse set of binary questions spanning macroeconomics, politics, technological developments, sports, and other miscellaneous events. For example, a representative question in ForecastBench asks:

Will an infrastructure disaster costing >$1B in a G20 country be widely attributed to an AI cyberattack before 2025?

Each event in these benchmarks resolves between July 2024 and June 2025, allowing us to observe the ground truth outcome (whether or not the event occurred).

Name	Questions	Resolution Range	Description
FB-7-21	498	7/28/2024 – 12/31/2050	Original Karger et al. (2024) benchmark.
FB-8-14	602	7/28/2024 – 12/31/2050	Updated Karger et al. (2024) benchmark.
FB-Market	76	7/25/2025 – 12/31/2050	Subset of Karger et al. (2024) sourced from prediction markets.
MarketLiquid	1610	4/2/2025 – 5/23/2025	More challenging benchmark sourced from liquid prediction markets.

Table 1: Overview of benchmarks used to evaluate the performance of the AIA Forecaster.

ForecastBench. The first variant of ForecastBench, which we call FB-7-21, is the 498-question benchmark previously studied in Karger et al. (2024). The authors report performance for a wide variety of LLM-based forecasters on this benchmark, as well as

3. Interestingly, we consistently found this class of mathematical techniques to be more robust than simple prompting changes, which we found largely ineffective at mitigating this problem.

the aggregate performance of human-generated forecasts in both general public surveys and surveys of expert superforecasters.[4] These questions are collected from a variety of public prediction markets and other established datasets which track world events. For additional detail on the data collection process we refer to Karger et al. (2024).

We also separately report performance on two variants of ForecastBench. The first, FB-Market, is the subset of FB-7-21 which is sourced from public prediction markets.[5] The second, which we call FB-8-14, is an updated snapshot of ForecastBench as of 08/14/2025 containing 602 questions.

MarketLiquid. Next, we curate a timely dataset of binary forecasting events sourced from a popular public prediction market platform. Our goal was to create a dataset such that (1) all of the markets resolve beyond the knowledge cutoff for the models considered in this manuscript and (2) includes events which are more relevant for policymakers and executives. For example, we restrict our attention to markets covering politics and policy, economics and financial markets, and developments in AI and technology; we exclude markets related to sports, weather prediction, and other miscellaneous categories which appear frequently in ForecastBench.

This dataset includes all markets which resolved between 4/2/2025 and 5/23/2025, but excludes markets which (1) were open for less than one week and (2) had fewer than 5,000 total contracts traded. This filtering yields 322 forecasting questions. For each question, we generate binary event forecasting questions at five dates spaced between the posted market open and market resolution date for a total of 1610 questions.[6] We provide additional detail on the curation of this benchmark, including the choice of dates and the prompts used to convert the raw data into a structured benchmark, in Appendix F.

Evaluation criteria. As is standard in the LLM forecasting literature (e.g., Halawi et al. (2024); Karger et al. (2024); Schoenegger et al. (2024)), we use the Brier score to evaluate forecasting performance (Brier, 1950) on each of the benchmarks described above. The Brier score for an individual forecast $p \in [0, 1]$ with ground truth outcome $o \in \{0, 1\}$ is simply the squared difference of the two, $(p - o)^2$. The Brier score for a set of events indexed by $i = 1, \ldots, n$ is the mean of this quantity, $\frac{1}{n}\sum_{i=1}^{n}(p_i - o_i)^2$. The Brier score ranges from 0 (perfect accuracy) to 1; a forecaster who uniformly predicts $p = 0.5$ for every event can achieve a baseline score of 0.25. The Brier score is known to be a strictly

4. Karger et al. (2024) also publish a larger "LLM-only" benchmark, which includes synthetically generated conjunctions of the original questions. We focus here on the benchmark which includes human performance.
5. As shown in Table 1, the FB-Market dataset includes questions with resolution dates as late as December 31, 2050. Additionally, a few questions are event-contingent and resolve immediately when the specified event occurs, provided it happens before a designated cutoff date (e.g., year 2100). For these questions, the resolution dates are indeed set at a far future date. Thus, the data set accommodates both fixed-date predictions and open-ended event forecasts.
6. Note that the market close date (the posted date by which the market will resolve) and the realized resolution date may be distinct, e.g., for events of the form "Will X happen by Y closing date" that occur strictly before the closing date. While we generate questions uniformly between the market open and *realized* resolution date, we only disclose the originally posted close date to the LLM forecaster.

proper scoring rule, which incentivizes truthful forecasting (see e.g., Gneiting and Raftery (2007) for an overview).

4.2 Forecaster Performance

In this section we present our headline results on FB-Market, FB-7-21, FB-8-14 and the MarketLiquid benchmarks. For FB-7-21, we include the performance of the following three baselines published in Karger et al. (2024): the state-of-the-art LLM forecaster on the Karger et al. (2024) leaderboard, the median forecast for each event in a general public survey, and the median forecast for each event in a survey of expert superforecasters.[7] We also include the performance of OpenAI's o3 reasoning model, which we find is the best off-the-shelf forecaster for which these questions do not appear in its pre-training corpus (see Appendix B for details). On FB-Market, we further evaluate performance against the market consensus (see Section 5 for details on prediction market mechanics). For FB-8-14, we again include the performance of OpenAI o3, the median of public surveys and superforecasters, and the best-performing LLM forecaster on ForecastBench. For MarketLiquid, we include the performance of the prediction market price as a baseline. These results are presented in Table 2 below.

Forecaster	FB-Market	FB-7-21	FB-8-14	MarketLiquid
Market	0.0965	×	×	**0.1106**
Public Survey (median per-event)	0.1035	0.1451	0.1510	×
Superforecasters (median per-event)	**0.0740**	0.1110	0.1152	×
ForecastBench state of the art	0.107	0.133	0.145	×
OpenAI o3	0.1096	0.1221	0.1262	0.1324
AIA Forecaster	0.0753	**0.1076**	**0.1099**	0.1258

Table 2: The performance of the AIA Forecaster on ForecastBench and MarketLiquid. The AIA Forecaster is statistically indistinguishable from expert superforecasters. Uncertainty quantification is provided in Appendix A.

In contrast to prior work, which shows that LLM forecasters dramatically underperform superforecasters, the AIA Forecaster is *statistically indistinguishable from superforecasters* on ForecastBench. Indeed, the point estimate for our system's performance is lower (i.e., more accurate) than the superforecaster baseline. The AIA Forecaster also substantially outperforms the o3 baseline, the general public survey, and the state-of-the-art LLM-based forecaster on the ForecastBench leaderboard.

7. We follow Karger et al. (2024) in using the median forecast for each event as the corresponding baseline for both superforecasters and the general public survey. This choice can be motivated by arguments which demonstrate that, under the assumption that individual forecasters make symmetric errors around the "true" forecast, the median is an unbiased estimate of this unknown value but the mean is not (Baron et al., 2014). Our empirical evidence corroborates this intuition: the per-event median of superforecasters (0.1110) modestly outperforms the per-event mean (0.1123), and the per-event median of the general public survey (0.1450) substantially outperforms the per-event mean (0.1586). The general public survey averages 49 responses per question (with a minimum of 40). The survey of superforecasters includes an average of eight responses per question (with a minimum of three).